

Cutflow table, normalized to 10.8 ab<sup>-1</sup>

| Process          | All         | 1lep 10<p<60 | ISO        | Veto extra iso p>20 | 10<E <sub>miss</sub> <55 | N=4+sqrt(d34)>14 | minNconst>8 | 85<mZ<95 | 40<pZ<60 | 120<mHcand<135 | 120<recoil<135 |
|------------------|-------------|--------------|------------|---------------------|--------------------------|------------------|-------------|----------|----------|----------------|----------------|
| ττ               | 50416483    | 17487786     | 16961795   | 15119079            | 529800                   | 224              | 0           | 0        | 0        | 0              | 0              |
| ZH(ZZ)           | 38569       | 8584         | 5432       | 3101                | 905                      | 776              | 550         | 332      | 156      | 128            | 121            |
| ZH(cc)           | 42575       | 5988         | 200        | 200                 | 124                      | 104              | 95          | 41       | 14       | 11             | 11             |
| ZH(gg)           | 120571      | 7271         | 614        | 614                 | 342                      | 313              | 300         | 144      | 45       | 33             | 30             |
| ZH(ττ)           | 92362       | 42554        | 37582      | 34859               | 9397                     | 2053             | 244         | 67       | 23       | 19             | 19             |
| ZH(bb)           | 857628      | 231386       | 14176      | 14163               | 9562                     | 8196             | 7483        | 4401     | 2353     | 2043           | 1971           |
| Z/γ→qq           | 594113065   | 40753274     | 2315367    | 2312430             | 369332                   | 56596            | 18546       | 7205     | 1890     | 1605           | 1581           |
| ZZ               | 14677092    | 3939305      | 2727222    | 705322              | 161611                   | 46489            | 21516       | 12190    | 2364     | 1800           | 1727           |
| WW               | 177535800   | 35342541     | 29570577   | 24432832            | 990492                   | 145636           | 22392       | 7557     | 2302     | 1784           | 1717           |
| ZH (H→WW)        | 314723      | 129895       | 106384     | 98406               | 61634                    | 48421            | 38409       | 24466    | 16681    | 15083          | 14727          |
| Total Signal     | 314722.8    | 129895.2     | 106384.5   | 98406.2             | 61634.2                  | 48421.1          | 38408.6     | 24466.2  | 16680.8  | 15083.4        | 14726.6        |
| Total Background | 837894145.0 | 97818687.9   | 51632965.2 | 42622600.1          | 2071564.9                | 260386.7         | 71126.2     | 31938.1  | 9147.0   | 7422.3         | 7176.1         |
| S/B              | 0.0004      | 0.0013       | 0.0021     | 0.0023              | 0.0298                   | 0.1860           | 0.5400      | 0.7660   | 1.8236   | 2.0322         | 2.0522         |
| S/sqrt(B)        | 10.9        | 13.1         | 14.8       | 15.1                | 42.8                     | 94.9             | 144.0       | 136.9    | 174.4    | 175.1          | 173.8          |